CleverLoan

Case Introduction

This example is drawn from an AI-assisted loan consulting system at a European retail bank, referred to as CleverLoan [Strich et al. 2012]. The bank's Small Loan Business Unit processed personal loan applications through certified loan consultants who autonomously advised customers, assessed creditworthiness, and made approval decisions. The unit introduced ClerverLoan, a supervised machine-learning system trained on historical loan data, to automate loan approval decisions and standardize the process.

In our models, Max represents an experienced loan consultant whose professional identity is grounded in autonomous decision-making, personal customer relationships, and expert judgment built over years of practice. Anna represents a junior consultant whose baseline role is limited to service desk tasks – welcoming customers, listening to needs, and assisting with pay-outs – and whose professional identity is aspirational rather than established. The Small Loan Business Unit is the organizational actor, and the Branch Manager occupies the Internal Verification Agent role, responsible for contract validation.

Unlike the Santander case, where the TA manager actively championed human-AI collaboration and designed identity-sustaining mechanisms, the Branch Manager in the CleverLoan setting is concerned primarily with operational efficiency rather than preserving consultants' professional identities. This difference in managerial orientation is central to the contrasting outcomes.

Identity Tensions

The introduction of CleverLoan generated divergent identity responses depending on baseline professional identity:

- For experienced consultants (Max): CleverLoan directly substituted the core tasks through which Max's professional identity was constructed – autonomous loan decisions, personal assessment of creditworthiness, and relationship-based discretion. This substitutive configuration threatened his self-concept as a "creative artist" and "problem solver" with authority over substantial financial decisions.

- For Junior consultants (Anna): The same system expanded professional scope by providing AI-generated guidelines that enabled participation in loan consulting – a domain previously inaccessible to her. CleverLoan augmented rather than substituted, because Anna's baseline identity was not anchored in autonomous decision-making.

This divergent response to the same AI system – threat for one, empowerment for the other – is the central analytical finding of this case and the reason it requires person-based modeling that distinguishes individual baseline professional identities.

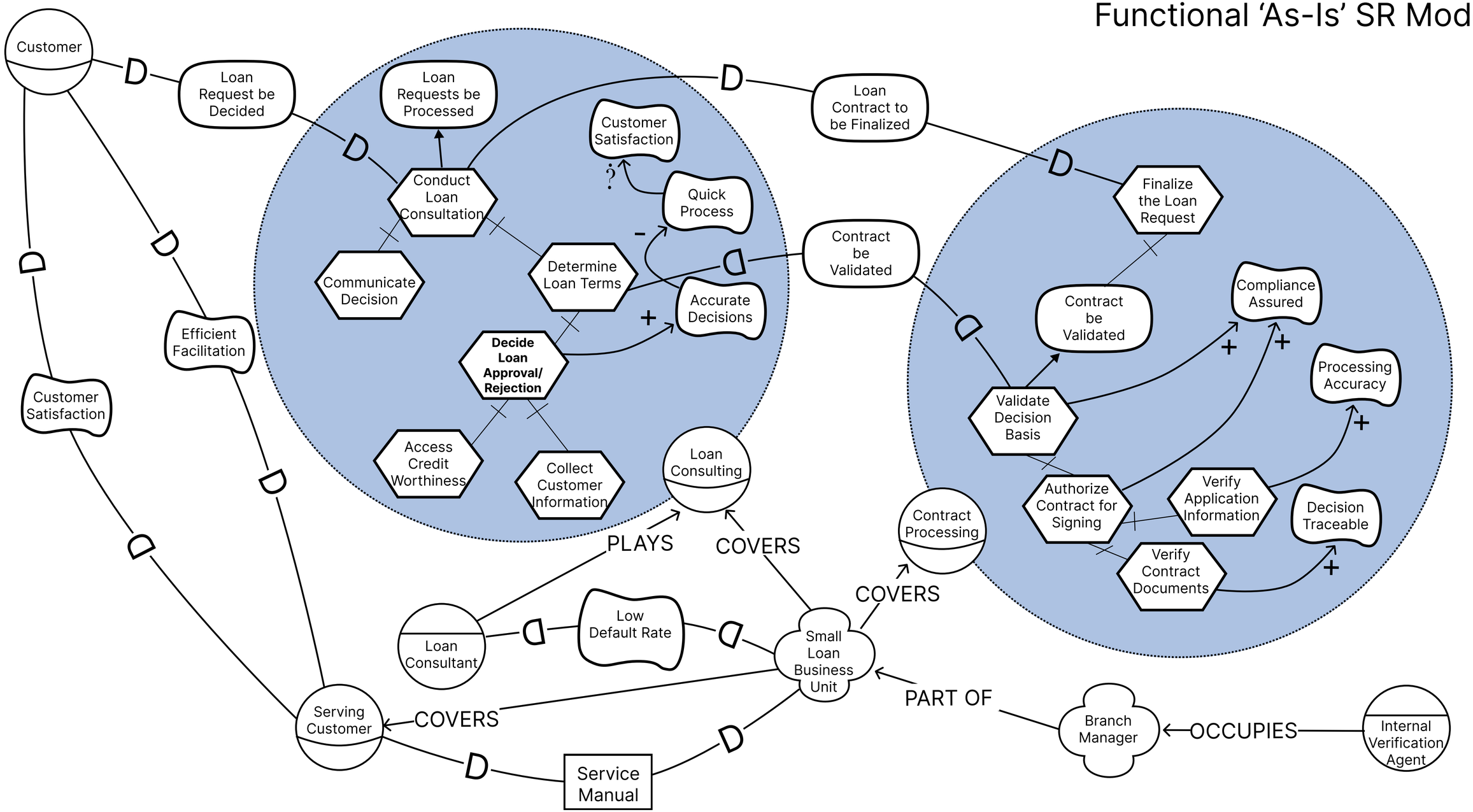
Functional 'As-Is' SR Model

What this model conveys:

- The loan consulting process is modeled using only role-based actors (Loan Consultant, Small Loan Business Unit, Branch Manager, Internal Verification Agent) and functional dependencies, without person-based identities.
- The Loan Consultant role PLAYS into the Loan Consulting boundary, which contains the full consultation workflow: Collect Customer Information, Access Credit Worthiness, Decide Loan Approval/Rejection, Determine Loan Terms, Communicate Decision, and Conduct Loan Consultation.
- Decide Loan Approval/Rejection is the identity-critical task – the primary locus of professional autonomy that CleverLoan will later automate.
- The Small Loan Business Unit COVERS both Loan Consulting and Contract Processing, with the Internal Verification Agent handling validation tasks (Validate Decision Basis, Verify Application Information, Verify Contract Documents, Authorize Contract for Signing).
- Softgoals at the role level – Quick Process, Accurate Decisions, Customer Satisfaction, Compliance Assured, Processing Accuracy, Decision Traceable – represent functional quality concerns only.

Key Analytical point: A purely functional analysis sees no reason not to automate Decide Loan Approval/Rejection: doing so would contribute positively to Quick Process and Accurate Decisions. The model does not capture that this task is constitutive of the loan consultant's professional identity, nor that experienced and junior consultants relate to this task in fundamentally different ways.

Functional 'As-Is' SR Model



Identity-Sensitive ‘As-Is’ SR Model

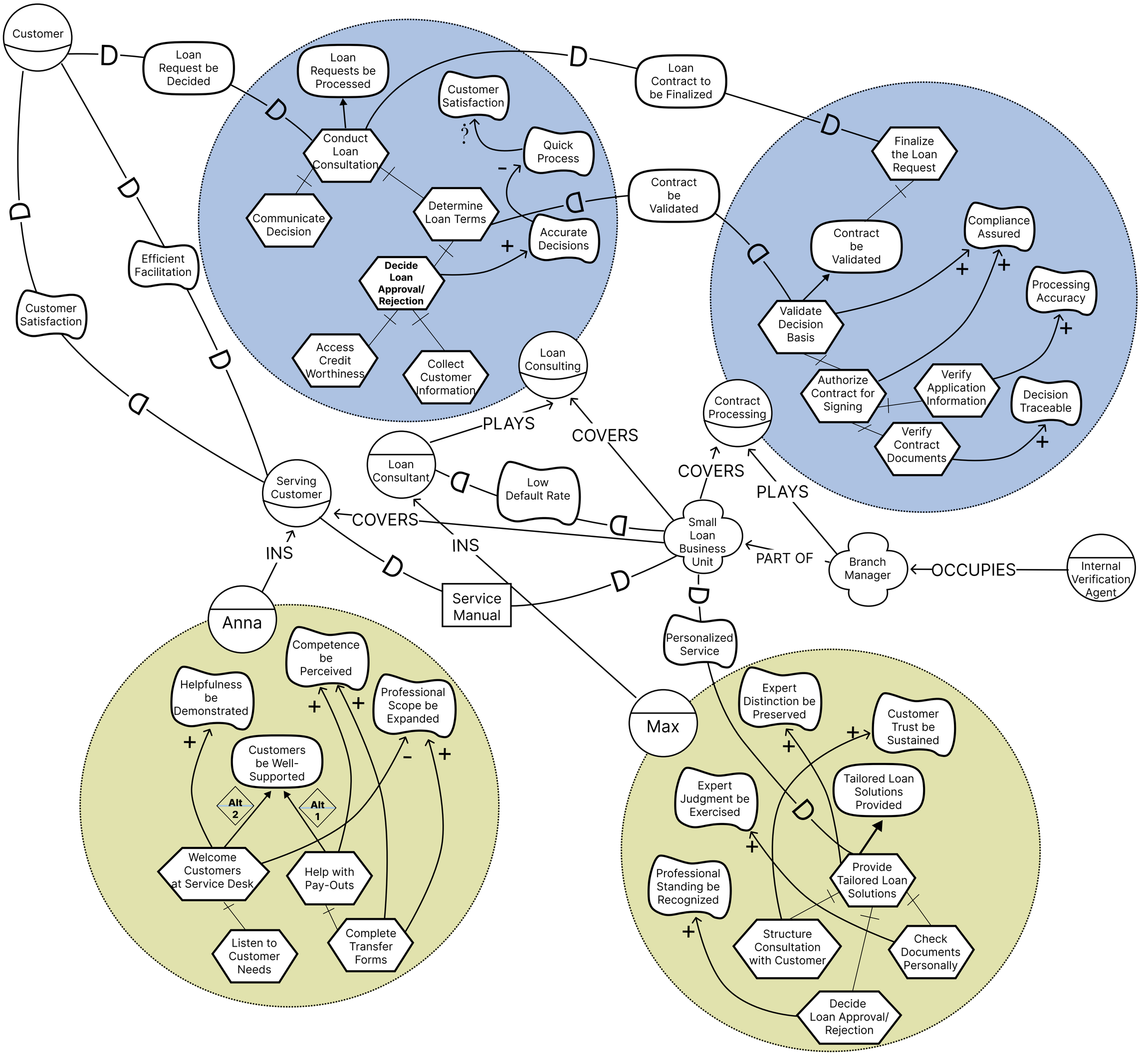
What this model conveys:

- Two person-based agents are introduced: Max (experienced consultant, lower right) and Anna (junior consultant, lower left), both instantiating the Loan Consultant role but with radically different baseline identities.
- Max's boundary contains task decomposition reflecting established expertise: Structure Consultation with Customer, Check Documents Personally, Decide Loan Approval/Rejection, and Provide Tailored Loan Solutions. His identity-sustaining softgoals — Expert Distinction be Preserved (+), Expert Judgment be Exercised (+), Professional Standing be Recognized (+), Customer Trust be Sustained — reflect a deeply anchored professional self-concept built through years of autonomous practice and sustained customer relationships.
- Anna's boundary shows limited scope with aspirational identity: her goal Customers be Well-Supported offers Alt 2 (Welcome Customers at Service Desk → Listen to Customer Needs) and Alt 1 (Help with Pay-Outs → Complete Transfer Forms). Her softgoals — Helpfulness be Demonstrated (+), Competence be Perceived (+), Professional Scope be Expanded (+) — reflect a professional identity that is developing rather than defending.
- The role-level boundaries (Loan Consulting, Contract Processing) are preserved from LC-1, showing that the functional workflow remains the same — what changes is the identity context.
- Anna additionally instantiates the Serving Customer role, while Max is associated with Personalized Service, differentiating how each enacts the consultant role in practice.

Key analytical point: The asymmetry between Max and Anna is the baseline that determines divergent AI responses. Max has everything to lose: his core tasks (Decide Loan Approval/Rejection, Provide Tailored Loan Solutions) are precisely what CleverLoan automates. Anna has everything to gain: her limited scope (Welcome Customers, Help with Pay-Outs) leaves room for AI-enabled expansion. This asymmetry — invisible in the functional model — is what identity-sensitive modeling makes explicit.

What this reveals beyond Santander: The Santander As-Is model had a single protagonist (Tim) whose identity was threatened. The CleverLoan As-Is model shows that the same organizational role can harbor fundamentally different person-based identities, producing opposite responses to the same AI system. This tests a capability

Identity-Sensitive 'As-Is' SR Model



of the modeling framework not exercised in the first case: representing experience-differentiated identity within a single role category.

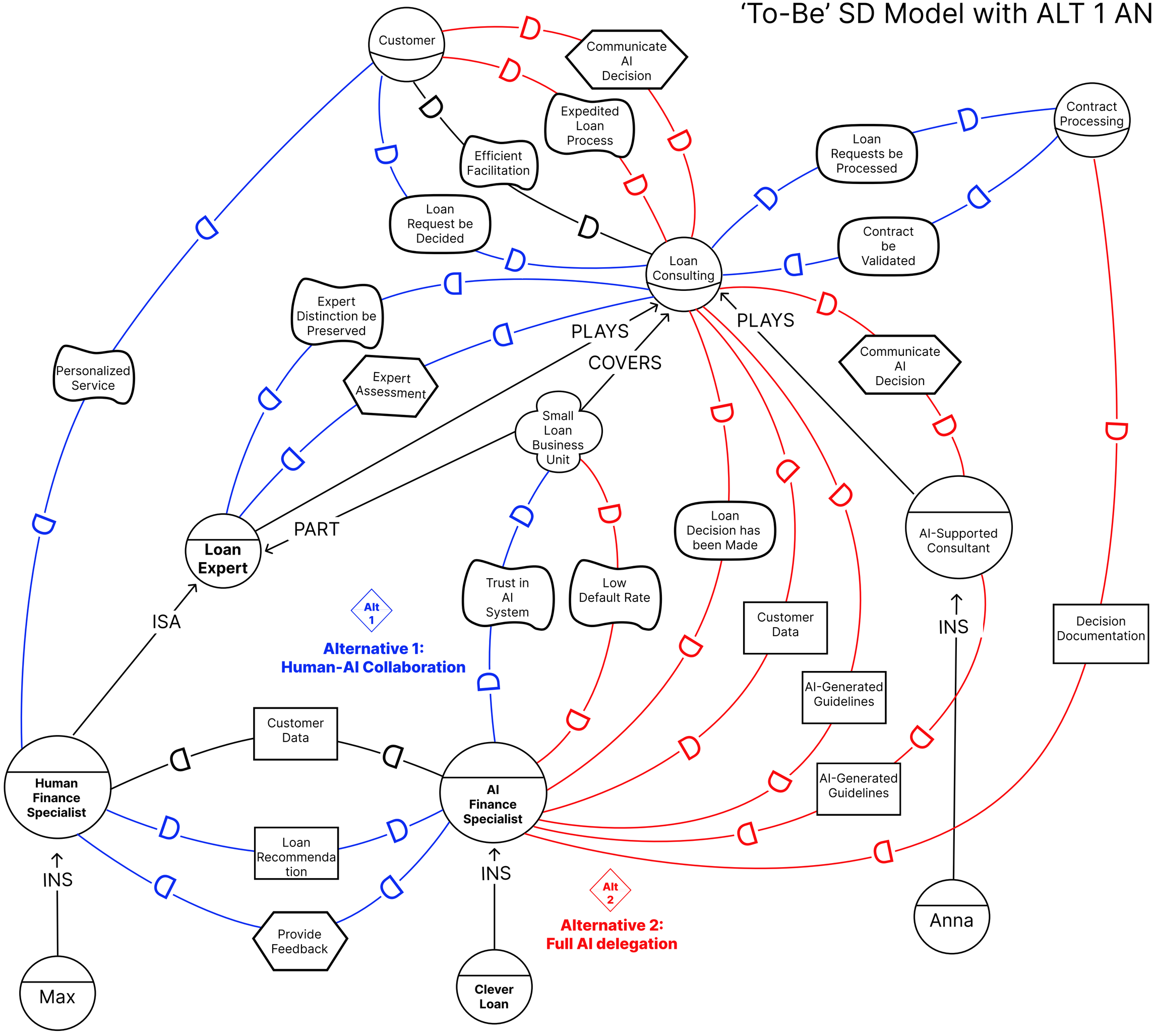
‘To-Be’ SD Model with ALT 1 and 2

What this model conveys:

- The Loan Expert role (which subsumes Loan Consultant in the redesigned organization) splits via ISA into Human Finance Specialist (instantiated by Max) and AI Finance Specialist (instantiated by CleverLoan), paralleling the Santander role split.
- A new role — AI-Supported Consultant — is created for Anna. This is not part of the ISA split; it is an entirely new organizational role enabled by AI, reflecting how AI creates professional identity pathways that did not previously exist.
- Two dependencies are shared across both alternatives (shown in black): Customer Data flows into the system (by humans) regardless of configuration, and the Customer depends on Efficient Facilitation in either case. These represent baseline organizational requirements that hold independent of identity-sensitive design choices.
- Alternative 1: Human-AI Collaboration (blue) shows reciprocal dependencies between Human Finance Specialist and AI Finance Specialist: Customer Data flows from Max to CleverLoan; Loan Recommendation flows back; Max provides Feedback to the AI. This reciprocal structure sustains relational identity through continued professional interaction.
- Alternative 2: Full AI delegation (red) routes through Anna as AI-Supported Consultant, receiving AI-Generated Guidelines and Decision Documentation from CleverLoan and communicating AI decisions to customers. Under this configuration, Max is excluded from the decision process.
- The Small Loan Business Unit COVERS Loan Consulting, with both the Human Finance Specialist (via Loan Expert) and the AI-Supported Consultant playing into it. Contract Processing handles finalization.
- Expert Distinction be Preserved appears as a softgoal associated with the Loan Expert role, and Trust in AI System appears at the organizational level.
- The Customer now receives Communicate AI Decision and depends on Expedited Loan Process and Efficient Facilitation.

Key analytical point: The two alternatives represent an organizational design choice: preserve the experienced consultant's role through human-AI collaboration (Alt 1), or maximize efficiency by delegating decisions to AI and using junior consultants to communicate outcomes (Alt 2). Unlike Santander, where the TA manager deliberately

‘To-Be’ SD Model with ALT 1 AND 2



chose Alt 1 to preserve identity, the CleverLoan setting — where the Branch Manager prioritizes efficiency — defaults toward Alt 2. Identity-sensitive modeling reveals what this choice costs.

What this reveals beyond Santander: The Santander To-Be SD split one role into Human/AI variants. The CleverLoan To-Be SD does this and creates an entirely new role (AI-Supported Consultant) that has no precedent in the As-Is model. This tests the framework's ability to represent AI-enabled role creation — not merely role redistribution — as an identity outcome.

‘To-Be’ SR Model with ALT 2

What this model conveys:

- The Loan Consulting boundary (blue) shows the post-AI workflow: Conduct Loan Consultation, Follow AI-Generated Guidelines, Understand AI Decision, with contributions to Quick Process (++), Expedited Loan Process (++), Accurate Decisions (+), and Customer Satisfaction (?/unknown).
- Max's boundary reveals the full complexity of identity threat when left unaddressed by organizational design:
    - Productive adaptation: Proactively Approach Customers Using AI Data (‘+’ Professional Standing, ‘+’ Customer Trust) and Provide Cross-Selling Advisory (‘+’ Professional Standing) show Max finding new ways to assert professional value using AI-generated insights.
    - Destructive resistance: Protect Professional Identity decomposes into Demonstrate Superiority Over AI-Dependent Consultants and Manipulate Data to Influence AI Decision, both contributing ‘- -’ to Compliance Assured and ‘- -’ to Organizational Cohesion. These are organizationally destructive behaviors that emerge specifically because identity threat was not addressed in the system's requirements.
    - Transparency gap: The softgoal AI Reasoning be Understood receives a ‘- -’ contribution, indicating that CleverLoan's black-box design Breaks the transparency condition that would enable Max to engage productively with AI reasoning. This directly contrasts with the Santander case, where AI reasoning Explained contributed ‘++’ to Professional Oversight Enabled. The same requirement category — transparency — surfaces in both cases, but with opposite outcomes: met in Santander (by design), broken in CleverLoan (by omission).

        - Communicate AI Decision fulfills the goal Customer be Served, while softgoals Specialist Distinction (-), Expert Judgment (-), and Customer Trust (-) reflect Max's discontinued professional self-concept.
    - Anna's boundary shows empowerment with dependency risk:
        - Follow AI Guidelines contributes (+) to Competence, (+) to Professional Scope be Expanded, and (+) to Helpfulness — genuine professional growth through AI augmentation.
        - Seek Reassurance From AI Provider contributes (-) to Competence, revealing that over-reliance on the AI system undermines the very competence that augmentation is supposed to build.
        - Authority loss: The softgoal Decision Authority be Preserved receives a (--) contribution from Refer Decision Responsibility to AI. This is analytically significant: Anna gains professional scope but never acquires decision authority. She transitions from having no authority (As-Is, service desk role) to operating under delegated authority that is not hers (To-Be, following AI guidelines). The requirement for authority preservation surfaces even for the empowered actor — revealing that scope expansion without decision authority is an incomplete form of professional identity development.
        - Enter Customer Data defines part of her operational scope; Customer Trust remains unknown (?), reflecting that the customer's response to AI-mediated decisions is not known.
        - The goal Customer be Served parallels Max's, but is achieved through fundamentally different means — compliance with AI guidelines rather than autonomous judgment.
    - The AI Finance Specialist (CleverLoan) processes Customer Data and produces Decision Documentation that flows to both Contract Processing and Anna's operational path.

Key analytical point: The “To-Be SR with ALT 2” does not propose a solution — it models the actual situation as it unfolded. Max's boundary shows that an unaddressed identity threat produces both productive adaptation and organizationally destructive behavior simultaneously. The same actor who proactively uses AI data for cross-selling also manipulates data to undermine the AI system's decisions. This duality — invisible without identity-sensitive modeling — represents a requirements gap: no functional requirement addresses the conditions under which experienced professionals turn from productive adaptation to destructive resistance. Anna's boundary shows the complementary risk: scope expansion without decision authority and with AI dependency creates an incomplete form of professional development that may itself become a future identity concern.

The two new softgoals — AI Reasoning be Understood (Max) and Decision Authority be Preserved (Anna) — ensure that all four requirement categories from the paper's Table 2 are exercisable in the CleverLoan models. Transparency and authority preservation surface as unmet requirements in the failure case, complementing the Santander case, where they were met through deliberate design. The framework thus demonstrates diagnostic value: it surfaces these requirement categories regardless of whether the organization addresses them.

What this reveals beyond Santander: The Santander To-Be SR showed how identity-sensitive design choices resolve tensions (transparency, authority preservation, professional growth). The CleverLoan “To-Be SR with ALT 2” shows what happens when such choices are absent: the framework can model not only solutions but also failure modes. Specifically:

- Transparency: Santander's "AI reasoning Explained" → (++) "Professional Oversight Enabled" versus CleverLoan's "AI Reasoning be Understood" ← (--) (black-box design). Same requirement, opposite outcomes.
- Authority preservation: Santander's "Decision Authority be Maintained" (met through collaborative design) versus CleverLoan's "Decision Authority be Preserved" ← (--) "Refer Decision Responsibility to AI" (unmet for both actors — lost by Max, never acquired by Anna).
- Competence: Santander's "Professional Competence" and "Professional Growth" (positive development) versus CleverLoan's mixed outcome (Anna: (+) from AI guidelines but (-) from over-reliance; Max: no development path at all).

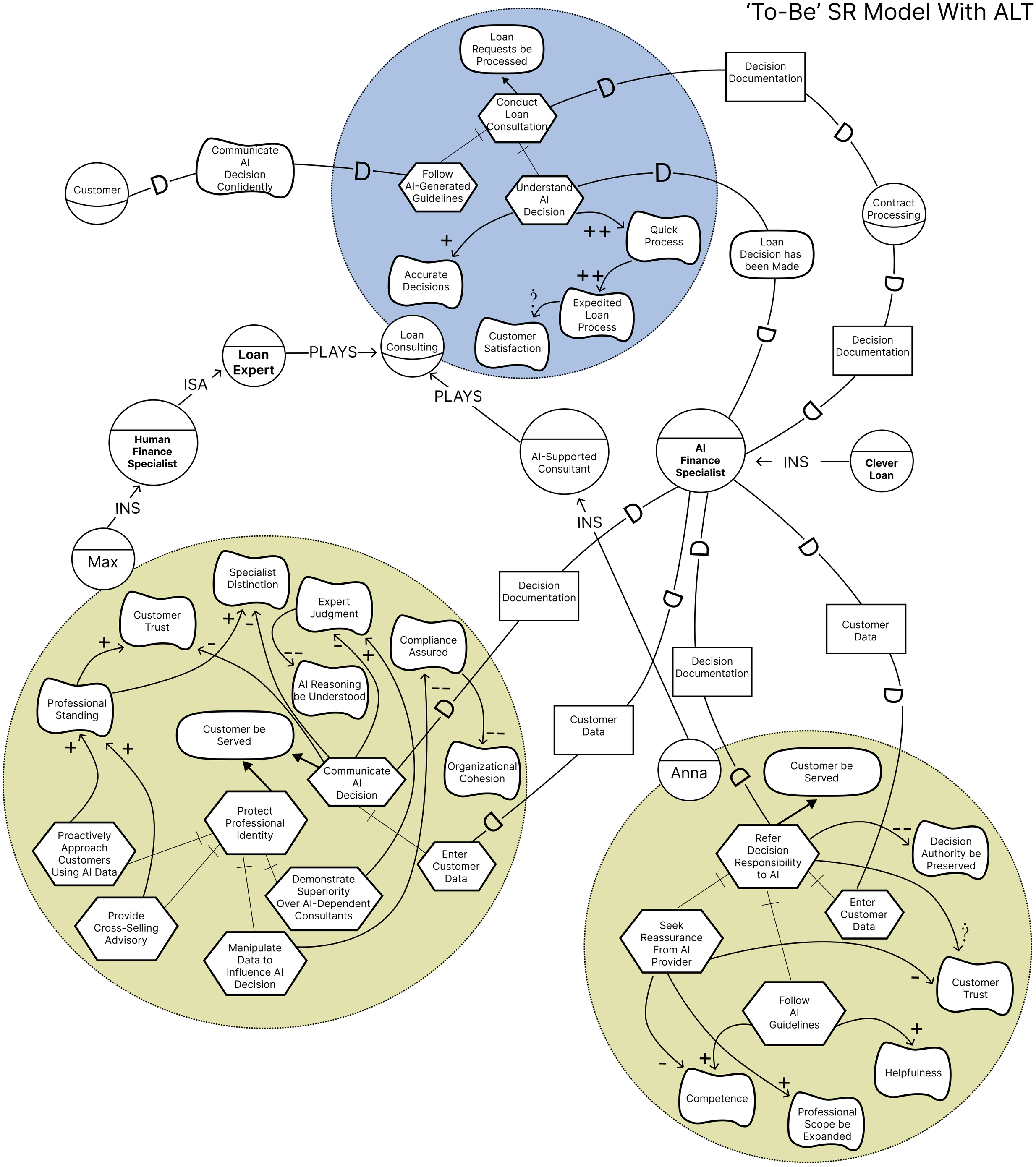
This tests the framework's diagnostic capability — its ability to surface identity-related requirements gaps by representing the consequences of ignoring identity in system design.

Coverage Analysis: What CleverLoan Adds to the Santander Case

Following the software testing analogy, the table below tracks which aspects of the identity-sensitive modeling framework are exercised by each case:

Modeling capability	Santander	CleverLoan	Notes

‘To-Be’ SR Model With ALT 2



Person-based identity (agent)	Tested (Tim)	Tested (Max, Anna)	Both cases exercise this
Role-based identity (role)	Tested (Hiring Expert, HR Professional)	Tested (Loan Consultant, Loan Expert)	Both cases
Relational identity (dependencies)	Tested (Tim–Sarah, Tim–AI)	Tested (Max–Customer, Anna–AI)	Both cases
Multiple agents in same role with divergent identities	Not tested	**Tested (Anna vs Max)**	New coverage
AI-enabled role creation (not just role split)	Not tested	**Tested (AI-Supported Consultant)**	New coverage
Identity-sustaining design (success path)	Tested (Alt 1 collaboration)	Not primary focus	Santander covers this
Identity threat without organizational mitigation (failure path)	Not tested	**Tested (Max's destructive behaviors)**	New coverage
Productive and destructive responses in same actor	Not tested	**Tested (Max adapts AND resists)**	New coverage
Empowerment with dependency risk	Not tested	**Tested (Anna's over-reliance)**	New coverage

Transparency as unmet requirement	Tested as met (AI reasoning Explained → ++ Professional Oversight)	**Tested as unmet (AI Reasoning be Understood ← −−)**	Same category, opposite outcome — new coverage
Authority preservation as unmet requirement	Tested as met (Decision Authority be Maintained)	**Tested as unmet (Decision Authority be Preserved ← −−)**	Same category, opposite outcome — new coverage
Managerial championing of identity preservation	Tested (Sarah)	Not present (Branch Manager prioritizes efficiency)	Contrasting managerial orientations
Contribution analysis on identity softgoals	Tested	Tested	Both cases
ISA role specialization for AI	Tested (Human/AI HR Expert)	Tested (Human/AI Finance Specialist)	Both cases
Reciprocal human-AI dependencies	Tested (feedback loop)	Tested (Customer Data / Loan Recommendation)	Both cases

Summary: CleverLoan adds seven new "test paths" not exercised by the Santander case: (1) divergent identities within the same role, (2) AI-enabled role creation, (3) unmitigated identity threat as a failure mode, (4) simultaneous productive and destructive responses in one actor, (5) empowerment with dependency risk, (6) transparency as an unmet requirement (diagnostic mode), and (7) authority preservation as an unmet requirement across both threatened and empowered actors. Together, the two cases provide broader coverage of the modeling framework's analytical capabilities, exercising all four requirement categories from Table 2 across both success (Santander) and failure (CleverLoan) conditions.

Domain-Specific Differences from Santander

Dimension	Santander	CleverLoan
AI type	Augmentative (CV screening support)	Substitutive (automated loan decisions)
Managerial orientation	TA manager championed human-AI collaboration	Branch Manager prioritized efficiency
Identity outcome	Resistance → role expansion (success)	Resistance + data manipulation (unresolved)
Stakeholder contrast	Single protagonist (Tim) with manager support (Sarah)	Two protagonists (Max, Anna) with divergent responses
Organizational structure	Hiring team with dedicated manager	Business unit with verification department
AI transparency	Real-time visualization and explanation	Black-box guidelines and documentation
Temporal trajectory	Four-year progression to collaboration	Ongoing tension without resolution